Investment and chronological operation of electricity storage and electrolysis

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Abstract

This study considers investment in batteries and electrolysers for the constructed energy system supplied in the project. It combines a deterministic investment appraisal (Tier 1) with a forecast-based chronological extension (Tier 2B). Before modelling, the five course datasets are reconciled to a common daily calendar. Missing generation records, query-endpoint snapshots, invalid demand observations and price-aggregation errors are repaired using archived upstream records, calibrated alternative series and documented imputation. Research on equipment costs and hydrogen support values provides a common financial basis.

Tier 1 uses mixed-integer optimisation to compare economic dispatch with supply-first operation over six historical years. Both policies choose electrolysis and zero batteries. A 10 t/day hydrogen outlet supports about 23 MW, whereas a national consumption ceiling supports hundreds to thousands of MW and illustrates the risk of using national demand to size a local asset. Replaying high-volatility periods and varying equipment costs and price dispersion tests when the investment would change.

The Tier 2B specification couples multi-step forecasts of generation, demand and electricity prices to rolling-horizon optimisation with storage-state feedback. It separates forecast evaluation from decision evaluation and fixes investment before the test period. This draft contains the completed Tier 1 analysis and the Tier 2B method and evaluation specification; forecast and control experiment results remain to be added.

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I. Introduction and data

1 Introduction

The client wants to know which flexible assets to add to its existing generation and how to operate them [1]. Tier 1 appraises batteries and electrolysers under historical market conditions. The Tier 2B specification considers operation when future prices, generation and demand are unknown.

We reconcile the five supplied datasets before modelling. The audit identifies missing and zero generation records, midnight observations used as daily averages, low demand readings and inactive price entries. We rebuild daily values from upstream observations where available and flag values estimated from other records.

Tier 1 combines market and equipment research with mixed-integer optimisation. We compare economic dispatch, which allocates electricity by value, with a rule that serves current electricity demand first. At an illustrative hydrogen offtake limit of 10 tonnes/day, both select 23.4 MW of electrolysis and zero batteries. Annual gains over the respective no-investment baselines are £12.68m and £2.44m. The analysis tests when these choices change with offtake, prices and equipment costs.

Tier 2B uses a rolling plan to apply only the current day's decisions and update the plan as new observations arrive. Forecast accuracy and operating value are evaluated separately in the design in Chapter III.

2 Data audit and reconstruction

The inputs are fuel-level generation, its reported total, a simulated wind farm, demand and market index prices. We align them on 2,452 UTC days from 1 January 2020 to 17 September 2026. Generation and demand are reported as mean MW and converted to daily MWh. Table I.1 summarises the repairs. Appendix A documents the evidence, methods and checks.

Table I.1. Data issues and treatments. Counts can overlap.

Issue	Treatment
Eight fuels recorded as zero (107 days)	Rebuild from Elexon generation and fuel outturn.
Solar recorded as zero (42 days)	Rebuild from Elexon or calibrated solar data.
Midnight values used as daily averages	Recalculate the mean over each full UTC day.
Dates absent from the original files	Restore source records and flag remaining estimates.
Missing or implausible demand readings	Fill invalid intervals with calibrated demand data.
Zero-volume price placeholders	Average active providers by volume, then by time.

The repaired series share a complete calendar. Fuel components reconcile to the total and the two price columns retain their factor-of-24 identity. Of the 2,452 dates, 1,929 meet the strict quality flag [3].

II. Tier 1: deterministic investment appraisal

1 Problem analysis

We select battery and electrolyser capacities and their daily use over 2020–25 (2,192 consecutive days). The investment is made once and applies to all six years. Direct electricity sales provide the no-investment benchmark. The investment case depends on three questions: whether electrolysis can use electricity that could otherwise be sold, how much hydrogen a buyer can take, and whether storage can shift enough electricity to recover its costs.

2 Research findings and model inputs

Table II.1. Research findings used in the model. Appendix B gives the evidence and calculations.

Research subject	Finding used in Tier 1
Electricity sales	No client-specific shortfall charge identified. Model uses market receipts.
Hydrogen value	UK support benchmark implies a scenario netback of £7.80/kg [8].
Hydrogen offtake	UK use implies a broad case of 1,559 t/day [11].
Battery equipment	Four short-duration units with 90–92% round-trip efficiency [12].
Electrolyser equipment	Eight units use 55.9–59.0 kWh/kg at rated load [12].
Equipment economics	Include battery capacity upkeep and electrolyser stack renewals [7, 12].
Price basis	Convert market prices and equipment costs to 2020 pounds [4, 9].

Table II.2. Per-unit equipment inputs. Capital and fixed operation are in 2020 pounds.

(a) Batteries

Model	Energy (MWh)	Power (MW)	RTE (%)	Capital (£k)	FOM (£k/yr)
ST255CS	0.257	0.125	90	65.6	1.90
LUNA215	0.215	0.108	90	55.1	1.60
LUNA241	0.241	0.108	90	60.8	1.76
Megapack 2XL	3.854	1.927	92	1,239	31.0

RTE is round-trip efficiency. FOM is fixed operation and maintenance. Use 20–80% state of charge and 15 years of service. The initial 20% fill is a commissioning cost. Main-case daily self-discharge is zero.

(b) Electrolysers

Model	Input (MW)	Output (kg/h)	Consumption (kWh/kg)	Min. load (%)
Neptune II	2.00	35	57	25
Neptune V	5.03	90	55.9	12.5
Alpha 50	50.31	900	55.9	12.5
MC250	1.31	22.125	59	10
MC500	2.54	44.25	57.3	10
MC1000	5.02	88.5	56.77	20
MC2000	10.05	177	56.77	20
MC4000	20.10	354	56.77	20

Use a 30-year life. Per MW, capital is £1.46m, fixed operation is £25.9k/year and each stack renewal costs £489.6k in years 11 and 22.

3 Notation and assumptions

Symbol	Definition
$t \in \mathcal{T},\ T,\ Y$	UTC day, number of days, represented years from calendar-year fractions.
G_t, D_t, S_t, p_t	Generation, market demand, surplus $\max(0, G_t - D_t)$, and price (£/MWh).
b_i, n_j, z_{jt}	Installed battery and electrolyser counts and online electrolyser count.
q_t, g_t^H, h_{jt}	Direct sales, direct input to electrolysis and input to model j.
$c_{it}, r_{it}, u_{it}, s_{it}$	Charge, discharge to sales, discharge to electrolysis and usable closing stock.
$W_i, P_i, \eta_i, \delta_i$	Usable capacity, power (MW), one-way efficiency ($\sqrt{\text{RTE}}$) and daily self-discharge.
H_j, α_j, κ_j	Full-load daily input per unit, minimum load fraction and specific use (MWh/kg).
v, w_j, L	Hydrogen netback and stack reserve (£/kg), and offtake limit (kg/day).
a_i^B, a_j^H	Annual costs per battery and electrolyser unit (£/year).

Energy and stock are in MWh. Indices i, j denote battery and electrolyser models.

Assumptions. Fixed generation combines the reported total and simulated wind farm. The single market caps sales at demand, with no imports, other exports, spatial limits, unused-power revenue or shortfall charge [1, 2]. A dispatches by value. B prioritises sales from generation, then storage, even at negative prices. Hydrogen is sold daily at £7.80/kg after non-electricity variable costs, capped at 1.56, 10 or 1,559 t/day. Costs use 2020 pounds, a 5% real discount and zero residual value.

4 Mathematical model

4.1 Objective

$$\max \quad \Pi = \frac{1}{Y} \sum_{t \in \mathcal{T}} \left[p_t \left(q_t + \sum_i r_{it} \right) + \sum_j (v - w_j) \frac{h_{jt}}{\kappa_j} \right] - \sum_i a_i^B b_i - \sum_j a_j^H n_j. \tag{II.1}$$

Annual cost a uses capital K, fixed operation F, life N, discount rate d and renewals C_k in year k.

$$a = F + \frac{d}{1 - (1+d)^{-N}} \left(K + \sum_k \frac{C_k}{(1+d)^k} \right). \tag{II.2}$$

Set $w_j = 0$ when renewals enter a. Gain is profit above the same-policy no-investment benchmark, so existing asset costs cancel.

4.2 Constraints

Daily constraints hold for all applicable t, i, j, with $T = |\mathcal{T}|$ and $(x)_+ = \max(0, x)$.

$$q_t + g_t^H + \sum_i c_{it} \le G_t, \quad q_t + \sum_i r_{it} \le D_t, \quad \sum_j h_{jt} = g_t^H + \sum_i u_{it}, \tag{II.3}$$

$$s_{it} = (1 - \delta_i) s_{i,t-1} + \eta_i c_{it} - (r_{it} + u_{it})/\eta_i, \quad 0 \le s_{it} \le W_i b_i, \quad s_{i0} = s_{iT} = 0, \tag{II.4}$$

$$c_{it} \le b_i \min(24 P_i, W_i/\eta_i), \quad r_{it} + u_{it} \le b_i \min(24 P_i, \eta_i W_i), \tag{II.5}$$

$$(c_{it} = 0) \ \vee \ (r_{it} + u_{it} = 0), \tag{II.6}$$

$$\alpha_j H_j z_{jt} \le h_{jt} \le H_j z_{jt}, \quad z_{jt} \le n_j, \quad \sum_j h_{jt}/\kappa_j \le L, \tag{II.7}$$

$$\text{Policy B:} \quad q_t = \min(G_t, D_t), \quad \sum_i r_{it} = \min \left\{ (D_t - G_t)_+, \ \sum_i \eta_i (1 - \delta_i) s_{i,t-1} \right\}. \tag{II.8}$$

Flows are nonnegative and $b_i, n_j, z_{jt} \in \mathbb{Z}_{\ge 0}$.

Constraint interpretation.

(II.3) allocates generation, caps sales and balances electrolysis inputs.

(II.4)–(II.6) carry stock across days with losses, limit throughput and allow one direction per battery model per day.

(II.7) imposes online loading and the daily offtake cap.

(II.8) gives B's remaining demand first claim on usable stock. The batteries can discharge that stock in one day. Binary indicators implement the direction choice and B's minimum.

Numerical checks. HiGHS certified 14 cases (relative gap below 10^{-9}). Independent checks covered flows, cash and dispatch, including six forced-storage cases [12].

5 Results and analysis

5.1 Capacity and historical performance

In the broad market case, electrolysis-only and joint investment return the same portfolios. A selects 3,631 MW (721 Neptune V and two Neptune II units) and B selects 740 MW (147 Neptune V units). At 10 t/day, both choose 23.4 MW across six smaller modules. Initial capital in the broad case is £5.30bn (A) and £1.08bn (B), producing about 488,000 and 19,100 t/year of hydrogen respectively.

Table II.3. Six-year portfolio choices and annual profit gains by policy. A dash marks an excluded investment. Gains are in £m/year.

(a) Allowed assets at the broad market cap

	Policy A			Policy B		
Allowed investment	Battery (MWh)	Electrolysis (MW)	Gain	Battery (MWh)	Electrolysis (MW)	Gain
Battery only	0	–	0	0	–	0
Electrolysis only	–	3,631	1,639	–	740	37.71
Joint choice	0	3,631	1,639	0	740	37.71

(b) Smaller outlets with both assets allowed

Daily offtake	Battery	Electrolysis (MW)		Gain	
(t/day)	(MWh), A/B	A	B	A	B
1.56	0	3.84	3.84	1.91	0.35
10	0	23.42	23.42	12.68	2.44

At 10 t/day, B loses money in 2021, 2023 and 2024, with a worst year of £1.26m below no investment. These years have 43, 29 and 30 surplus days, respectively, against 154 in 2022. A can produce on additional days when the hydrogen value exceeds the price of forgone electricity sales. At the broad cap, A meets 82.2% of electricity demand and B meets 92.2%. Using the alternative UK cost basis at that cap leaves A's capacity unchanged and raises B's from 740 to 1,465 MW, with zero batteries in both cases [7, 12].

5.2 Energy attribution and contract sensitivity

The historical mean physical surplus is 2,276 GWh/year. For the zero-battery solutions we attribute $\min(g_t^H, S_t)$ to physical surplus first. The remaining electrolysis input uses energy within current electricity demand. We calculate the monetary loss separately against same-policy electricity sales without investment, accounting for negative prices under A. The aggregate records identify the available fuels by day and cannot trace the generating unit that supplied electrolysis.

Table II.4. Electrolysis sourcing and forgone electricity receipts. Energy is in GWh/year. The revenue reduction is in £m/year.

Offtake	Policy	Total input	Physical surplus	Within current demand	Electricity revenue reduction
Broad	A	27,255	2,184	25,070	1,618
Broad	B	1,067	1,067	0	0
10 t/day	A	183	43	140	9.18
10 t/day	B	43	43	0	0

In the broad market case, A uses about 2,184 GWh/year of physical surplus, only 8% of its total electrolysis input. Its electricity receipts fall by £1,618m/year. B uses 1,067 GWh/year of surplus and preserves direct sales, limiting its hydrogen throughput. At 10 t/day, each captures about 43 GWh/year, just under 2% of the physical surplus.

After reoptimising electrolysis and dispatch, the lowest netbacks that support any investment are about £3.32/kg in A and £4.57/kg in B at 10 t/day. At 1.56 t/day they rise to £3.38/kg and £4.66/kg. The 23.4 MW portfolio in Table II.3 is sized at £7.80/kg. Holding its operation fixed, it breaks even at £3.90/kg in A or £4.61/kg in B. At other prices, reoptimise operation and capacity.

5.3 Storage and investment sensitivity

Table II.5. Storage entry under historical and stressed conditions. Price multipliers scale the standard deviation of daily prices.

Test	Implication for storage investment
Historical high-volatility windows	All four battery options remain unselected in the highest-variance 60-, 90-, 180- and 365-day windows under both policies. The 90-day price standard deviation is £114/MWh against £65/MWh over six years.
Battery cost and price stress	Halving purchase and maintenance costs still gives zero batteries. With historical supply and demand, batteries first enter at about 40× (A) and 127× (B) the historical price standard deviation at the modelled costs.
Joint supply and price stress	The joint grid selects zero batteries at the historical price dispersion across nine supply–demand gap dispersions. At 120× price dispersion in B, raising gap dispersion from 1× to 1.1× changes the optimum from zero to 2.60 GWh.
Cavern compressed air	A separate 50 MW / 1.5 GWh test, at one-third of the reference battery installation cost per MWh and 62% round-trip efficiency, needs 5.38–5.50× (A) or 15.00–15.13× (B) price dispersion at the original supply–demand gap.

The volatility tests centre each series before scaling, preserving means and date order. The battery and cavern tests allow the electrolyser choice to adjust. For battery-only investment, even free charging and the full usable discharge on every positive-price day yield at most 57% of annual ownership cost across the four models. At historical prices the cavern's annual operating gain is £1.38m in A and £2.35m in B, against an annual cost of £8.52m [13].

5.4 Investment recommendations

Size electrolysis to secured demand. For 10 t/day, appraise the 23.4 MW portfolio against the £34.2m initial cost. A and B yield £12.68m and £2.44m/year respectively (Table II.3). Confirm the ability to displace electricity sales before relying on A. Under B, use the observed loss years to test cash reserves and the case for staged procurement. Reduce the appraisal to roughly 3.84 MW if the outlet is near 1.56 t/day. At a netback below the market entry levels above, defer electrolysis at the tested costs. The broad cap describes market scale. Capacity at that scale warrants appraisal only with much larger contracted offtake.

Defer daily storage. Both cost and historical stress tests support zero batteries (Table II.5). Reopen procurement if a lower installed cost or additional revenue brings the expected operating gain above annual ownership costs. For the cavern, the historical gain implies an annual cost ceiling of £1.38m under A or £2.35m under B on these inputs.

III. Tier 2B: forecasting and chronological operation

1 Information and experimental design

The operating decision is made once per day. Following the classroom simplification, current-day generation, demand and price are known at that decision, while future days are unknown [2]. Let $y_t = (G_t, D_t, p_t)$ and let $\mathcal{I}_t$ contain observations through day t, the installed equipment and the previous day's usable stock. Only information in $\mathcal{I}_t$ can determine the implemented action. Current-day aggregates are treated as observed at this daily planning decision.

We reserve 2020–22 for training and capacity selection, 2023 for model and horizon selection, and 2024–25 for the final test. A forecasting training target must lie before its corresponding cutoff. Repair calibration, scaling and hyperparameter selection use training/validation data only. The six-year Tier 1 capacity comparison remains a hindsight benchmark; test observations must not choose the capacity used in the predictive experiment.

2 Multi-step forecasting model

The proposed first predictive model is regularised linear autoregression, compared against a seven-day seasonal-naive predictor. At each forecast origin t, the feature vector f_t contains observed generation, demand and price lags, day-of-week indicators and annual sine/cosine terms. For variable $j \in \{G, D, p\}$ and horizon $k = 1, \ldots, H$, fit

$$\widehat{\beta}_{jk} = \arg\min_{\beta} \sum_{\tau \in \mathcal{A}_k} (y_{\tau+k,j} - f_\tau^\top \beta)^2 + \lambda \|\beta\|_2^2, \qquad \widehat{y}_{t+k,j} = f_t^\top \widehat{\beta}_{jk}, \tag{III.1}$$

where $\mathcal{A}_k$ contains eligible training origins whose targets are already observed, $\lambda \geq 0$ is the validation-selected penalty and H is the planning horizon. Forecast generation and demand are projected onto nonnegative values; negative price forecasts remain allowed. The seasonal-naive forecast uses the most recent observation with the same weekday, recursively if needed.

3 Rolling optimisation and storage feedback

At day t, installed battery and electrolyser counts are fixed. A planning problem uses observed values for the current day and predictions for $t+1, \ldots, t+H$. It retains the Tier 1 flow, demand, throughput and electrolysis constraints, but replaces the pooled energy balance with chronological stock:

$$s_\tau = s_{\tau-1} + \eta c_\tau - r_\tau/\eta, \qquad 0 \leq s_\tau \leq E_B b, \qquad \tau = t, \ldots, t+H. \tag{III.2}$$

Here s_{t-1} is the measured initial stock for that solve. The optimiser maximises forecast operating receipts plus a terminal-stock value; the latter is selected on validation data to avoid repeatedly liquidating storage at an artificial short horizon. Only (q_t, c_t, r_t, h_t) is implemented. Its realised stock becomes the next solve's initial condition. The test's initial and final stock treatment is held equal across all compared policies.

4 Evaluation and results to be completed

Forecast error is reported by series and horizon using mean absolute error and root mean square error, with price errors in £/MWh and physical errors in MWh. Operating evaluation compares the predictive controller, the same controller with seasonal-naive forecasts, a current-day rule and a full-path perfect-information optimiser, all at identical installed capacities and with identical data, costs and terminal stock. The gap to perfect information measures the realised loss from restricted information, rather than a change in investment scale.

Since zero batteries remove cross-day storage, fixed nonzero battery cases will diagnose prediction and feedback value alongside the training-selected investment. Results will include forecast errors, realised annual value, surplus capture, lost sales, stock feasibility and seasonal/adverse-period comparisons. These forecast and control experiments are pending.

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A. Data reconstruction

1 Sources and alignment

We archive the five course inputs and align their date keys on a continuous UTC calendar from 1 January 2020 to 17 September 2026. The original fuel total is checked against the fuel breakdown. When the fuel data are reconstructed, their sum replaces the reported total. The wind-farm file is a simulation. We retain its ten zero-output days and exclude its 18 September 2026 row from the common calendar. Within that calendar the simulation values are preserved. The original files omit 30 generation dates, one demand date and 30 price dates. The repairs use Elexon's AGPT, FUELHH, VD and MID archives, and NESO historical demand and embedded generation [3].

2 Generation

Missing and zero values. The simultaneous zeros in eight fuel categories on 107 course dates are inconsistent with contemporaneous Elexon generation records. We use complete Elexon Actual Generation per Production Type (AGPT) data where available and reconstruct the eight categories from Elexon half-hourly fuel outturn (FUELHH) where needed. Gas combines combined- and open-cycle generation. Pumped storage counts positive output only. Other coincident partial zeros are flagged separately from normal fuel-specific zeros. For the 42 zero-solar dates, we prefer complete AGPT records, interpolate short gaps and estimate heavily missing days from calibrated National Energy System Operator (NESO) embedded solar data. One date belongs to both zero-value groups. Validation checks that all 107 fuel dates and 42 solar dates were repaired.

Incomplete coverage and export differences. Seven course days reproduce the midnight query endpoint. We reaggregate the complete UTC days. The current AGPT archive contains 264 incomplete days, and 136 days with simultaneous zero periods. For short gaps of at most six half-hours, we interpolate from valid observations on the same day, using the nearest observation at a day boundary. For larger gaps, the eight non-wind/solar categories use FUELHH and renewable components use calibrated alternative data. Solar is scaled to AGPT by year. Wind uses a local fit based on FUELHH wind and NESO embedded wind, with the nearest 60 complete training days in the same year and every fifth day held out. We rebuild the 87 differences between the 2026 extract and the current source, including 61 dates outside the initial 107-date repair.

3 Demand and market index prices

Demand. The original demand data reproduce Elexon demand outturn (VD) query endpoints on 349 Tuesdays and the final file date. Requerying the full day removes this sampling error. The retrieved VD output has a five-minute grid, with 116 days containing fewer than 288 records and 42 days having readings below 10 GW. We keep valid readings and fill gaps with contemporaneous NESO transmission system demand plus a local median VD-to-NESO offset. Nearby complete days supply the offset on two dates with only one VD record. The method reconstructs 153 days. On 2 August 2022 the upstream feed has 17 consecutive readings at 636 MW. Replacing those intervals and averaging the full day yields an estimated 28,744 MW.

Prices. Elexon Market Index Data (MID) reproduce the 346 query-end prices and the remaining 2,076 published daily values under the original rule. We remove 115,571 provider records with zero reported traded volume, then volume-weight active providers within each half-hour and average those half-hours over the UTC day. Genuine zero or negative prices with traded volume are retained. On 1 January 2020 the daily price changes from £18.59 to £37.19/MWh. Seven nominally negative course dates become positive on complete-day averaging. The repaired series has two negative daily means. Short gaps are interpolated. For 35 missing or severely incomplete price days, estimates use four nearby observed days with the same weekday. We maintain both price columns for compatibility and use the £/MWh series in the model.

4 Repair outcomes and verification

The release contains 2,452 observations per series, a cell-level record of original and replacement values, and flags for the method used. Across the calendar, 436 dates use an estimate or interpolation, 323 use cross-source reconstruction, and 1,929 meet the strict quality flag. The categories can overlap. Independent checks reconcile fuel components to total generation, verify the price-column factor of 24 and recompute 2,372 prices with complete active-provider coverage.

Held-out comparisons measure the uncertainty of reconstruction. Annual solar mean absolute error ranges from 42 to 182 MW, onshore wind from 363 to 592 MW and offshore wind from 453 to 832 MW. Hiding the first 17 demand intervals on 74 control days gives a daily mean absolute error of 8.63 MW. Hiding a whole price day on 77 control days gives £19.87/MWh mean absolute error. These validation errors describe the held-out days. The archived inputs, evidence, audit and validation records are identified in [3].

B. Market and equipment research

1 Electricity trading and hydrogen offtake

Electricity. The course demand series represents a constructed market. Published imbalance settlement applies to differences between contracted and metered positions. A company-specific charge would require an actual delivery commitment, settlement terms and an applicable price. We model sale at the daily market price and report demand coverage separately. Policy B is a dispatch scenario for testing priority.

Hydrogen price. The sources examined provide distinct measures. EEX HYDRIX is a weekly German price indication including delivery. ICIS publishes assessed values linked to production costs and power arrangements. Argus provides modelled costs, including UK proxies. The UK Hydrogen Allocation Round 1 (HAR1) reports a supported strike price, while the Low Carbon Hydrogen Agreement distinguishes the buyer's achieved sales price from support payments. We use the HAR1 reference as a constant netback scenario. At the benchmark of £175/MWh in 2012 prices and 39.4 kWh/kg, the converted value is $175(39.4/1000)(108.7/96.1) = 7.799$ in 2020 £/kg. The scenario applies no additional non-electricity deduction and assumes the buyer accepts each model's gas quality [8, 4, 9, 12].

Hydrogen volume. The European Hydrogen Observatory's 2025 update reports about 569,024, 474,414 and 485,626 tonnes of UK consumption in 2022, 2023 and 2024. We divide by the respective calendar days to obtain approximately 1,559, 1,300 and 1,327 t/day. The largest is a broad market case that smooths aggregate consumption across the year and includes existing captive use. The 1.56 t/day case is 0.1% of that benchmark and 10 t/day is an illustrative buyer case. Actual offtake depends on location, purity, delivery capacity and contract terms [11, 12].

2 Battery selection and cost basis

Sungrow ST255CS-2H, Huawei LUNA215 and LUNA241 and Tesla Megapack 2XL are compatible with the two-hour reference. We use model-specific power, nominal capacity and AC round-trip efficiency, splitting the latter symmetrically between charge and discharge. The 20–80% operating band is a common modelling convention. It provides 60% of nominal capacity as usable stock. In the main case, stated AC efficiency covers round-trip energy losses and we apply zero explicit daily self-discharge [1, 5, 12].

For the three smaller units, the Danish Energy Agency installed-cost reference is scaled by energy capacity and converted to 2020 pounds. The fixed annual budget is 2.9% of capital. We also test the catalogue's literal capacity-based operating-cost line as an alternative. For Megapack, the 2024 US public estimate is US$9.76m installed for five 2XL units. The model transfers the average price of US$1.95m per unit to single-unit decisions at 2020 exchange and price equivalence. The 2026 equipment-only listing omits installation and tax. For a lower-cost case we add a Lazard-based installation proxy. The Lazard cost boundaries distinguish standard maintenance, warranty and capacity augmentation. We adopt the US National Renewable Energy Laboratory 2.5%-of-capital fixed annual budget for the maintained-capacity case. It includes the upkeep provision, so no separate augmentation bill is added. The historical US estimate provides a scenario price for the UK project. The available sources do not establish a bulk-discount curve [5, 9, 12].

3 Electrolyser selection and cost basis

The eight models in Table II.2 use published manufacturer ratings. We obtain actual input MW from the rated hourly output and specific electricity consumption and use the product's minimum sustained load. Equipment is selected in whole units. The wet-gas POSEIDON reference would require a matching treatment and delivery cost for comparison with the selected product grades [1, 12].

For the main comparable case, the midpoint of the International Energy Agency's 2024 non-China installation range of US$2,000–2,600/kW is converted to £1.46m per MW in 2020 prices. The UK production cost study supplies fixed maintenance and absolute stack replacement costs, normalised to AC input power. The alternative uses the UK cost study's capital and fixed-operation package with a variable replacement reserve per kg, in place of the scheduled replacement charges [6, 7, 9, 12].

4 Financial conversion and operating horizon

Monthly nominal electricity prices are multiplied by the ratio of the 2020 mean Consumer Prices Index to the index in the observation month. The course battery reference of EUR 288,000/MWh corresponds to £256,235/MWh at the 2020 exchange rate. The comparable US equipment conversion factor is 0.635 in 2020 pounds per 2024 dollar. Annual equivalent costs use a 5% real discount rate, 15 years for batteries and 30 years for electrolysers, with zero residual value. The commissioning charge purchases the starting 20% battery fill at £78.13/MWh, allowing for charge efficiency. Subsequent stock is measured above that level and returns to it at the end of the study. Electricity infrastructure costs that are identical across investments cancel against each policy's no-investment comparator [4, 9, 12].